

Trainings ▾

General Information

Follow the Normal Starting Procedures in this section in cold weather. The Wait-To-Start lamp will stay on longer.

With Mechanical or Electrical Metering Equipment (Ether)

- Disengage the driven unit, or if equipped, put the transmission in neutral
- While cranking the engine, inject metered amounts of starting fluid
- Engine oil pressure **must** be indicated on the gauge within 15 seconds after starting.

Ether Starting Aids

WARNING

Because of the potential for an explosion, do not use volatile cold starting aids in underground mine or tunnel operations. Ask the local U.S. Bureau of Mines inspector for instructions.

WARNING

Starting fluid is highly flammable and explosive. Keep flames, sparks, and arcing switches away from starting fluid.

WARNING

Do not breathe starting fluid fumes. Starting fluid fumes can be hazardous to your health.

CAUTION

Do not use excessive amounts of starting fluid when starting an engine. The use of too much starting fluid will cause damage to the engine.

- Spray starting fluid into the air cleaner intake while another person cranks the engine.

Grid Heater

WARNING

To reduce the possibility of personal injury and property damage, never use starting fluid if the grid heater option is used. Starting fluid, which contains ether, can cause an explosion.

A grid heater is available that improves cold weather starting characteristics by heating the intake air. It can also serve to reduce white smoke if it is energized during cold ambient temperatures while the engine is at idle.

The electric grid heater operates in a preheat and postheat mode. The length of heater on-time is a function of the engine temperature.

If the intake manifold air temperature is greater than 19°C [66°F], the electric grid air heater system will **not** be activated. If the intake manifold air temperature is below 19°C [66°F], the system will operate as follows:

Engine Starting Cycle

1. Turn the ignition key to the RUN position. When the key is in this position, the WAIT-TO-START lamp will be illuminated for approximately 25 seconds. Do **not** crank the engine until the WAIT-TO-START lamp shuts off.

Note : The controller is reset each time the ignition is turned off and the cycle will start over.

2. When the WAIT-TO-START lamp goes out, the preheat cycle is complete. Crank the engine.

Postheat Cycle

3. Postheating occurs as the grid heater elements are cycled for a while with the engine running. Postheating helps warm the engine up faster and eliminates white smoke. Postheating is determined by the intake manifold air temperature upon start-up.